

MDI3003 — ADVANCED PREDICTIVE ANALYTICS

House Price Prediction using Linear Regression

From Statistical Baseline to an End-to-End Reproducible Regression Study

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Abstract

Automated valuation is one of the oldest and still one of the most commercially consequential applications of regression analysis. This report documents a complete, reproducible regression study that predicts residential property value from measurable structural, locational, and neighbourhood attributes, built around Linear Regression as the interpretability anchor and extended with regularized, tree-based, and ensemble comparisons.

Two datasets of deliberately different character were used. Dataset A is the California Housing collection (20,640 observations, eight numeric predictors, target expressed as median district house value in units of \$100,000), which offers a clean, fully numeric, large-sample setting dominated by income and geography. Dataset B is the Ames Housing dataset (1,460 residential sales, 79 usable predictors spanning numeric, ordinal, and nominal fields, target SalePrice in US dollars), which introduces realistic missingness, high-cardinality categorical structure, and a right-skewed price target more representative of an individual-transaction appraisal problem.

On Dataset A, a tuned Random Forest Regressor produced the lowest test-set error (RMSE 0.5038, R^2 0.8063 against a mean-baseline RMSE of 1.1449), while ordinary Linear Regression established an interpretable floor (RMSE 0.7456, R^2 0.5758). On Dataset B, Gradient Boosting achieved the strongest raw test performance (RMSE 27,183, R^2 0.9037), while a log-target Ridge Regression — an explicit ablation on the skewed SalePrice target — reached a competitive RMSE of 25,726 with full linear-model interpretability retained. Across both datasets, five or more models were compared under an identical, leakage-safe preprocessing pipeline with a shared five-fold cross-validation protocol, and at least one regularized model was tuned via grid search.

The study finds that non-linear ensembles consistently out-predict the linear baseline in raw error terms, but the margin is not uniform: it is comparatively modest on the geography-and-income-driven California data and considerably larger on the feature-rich, non-linear Ames data, where overall build quality alone accounted for the majority of Random Forest feature importance. The report closes with a discussion of residual behaviour, segment-level error patterns, responsible-use considerations for location-sensitive valuation models, and a deployment-oriented recommendation that favours a tuned, log-transformed regularized linear model as the operational default, with the ensemble retained as a benchmarking and anomaly-flagging companion rather than the sole production model.

Introduction

Valuation is the point where a housing market becomes computable. Before any transaction, mortgage decision, or tax assessment can proceed at scale, someone has to turn a bundle of physical and locational attributes — square footage, age, room counts, neighbourhood — into a single defensible number. For most of the twentieth century this was a manual, comparables-driven exercise performed by licensed appraisers. Regression analysis, and later machine-learning regression, offered a way to make that mapping explicit, repeatable, and auditable, which is precisely why house-price prediction has become the canonical entry point for teaching supervised regression in both academic and industry settings.

This report is the technical deliverable for MDI3003 Laboratory 01, which uses the house-price problem to move a student from a single-feature statistical baseline to a fully compared, tuned, and diagnosed regression pipeline. The work is organised in two tiers, mirroring the laboratory's own two-tier design: a core exercise that establishes a defensible Linear Regression baseline on one dataset, and an extended study that repeats the workflow on a

second, structurally different dataset and adds regularized, tree-based, and ensemble comparisons, cross-validation, hyperparameter tuning, and residual diagnostics.

Everything reported here — every metric, coefficient, figure, and table — is taken directly from two executed Jupyter notebooks (one per dataset) submitted alongside this report. No result has been altered, rounded upward, or invented for presentation purposes; where a number looks unusual (for instance, a negative Adjusted R^2 on Dataset B), the report explains the statistical reason rather than silently correcting it.

Lab Objectives

The laboratory and its extended study were carried out against the following objectives, each of which is addressed with concrete evidence later in this report:

- Translate the real-estate valuation need into a supervised regression problem with an explicit target, observation unit, and success criterion.
- Build simple and multiple Linear Regression models and interpret their coefficients in property-pricing terms.
- Construct a leakage-safe preprocessing pipeline that treats numerical and categorical fields correctly and is fitted only on training data.
- Evaluate predictive performance with MAE, RMSE, R^2 , Adjusted R^2 , cross-validation, and residual diagnostics.
- Compare Linear Regression against regularized (Ridge, Lasso, Elastic Net), tree-based (Decision Tree), and ensemble (Random Forest, Gradient Boosting) regressors.
- Document assumptions, decisions, limitations, and deployment risk in an industry-style technical report.

Learning Outcomes

Table 15 at the end of this report cross-references each rubric criterion to the section that satisfies it. The eight learning outcomes targeted by the laboratory, and how this study demonstrates them, are summarised below.

Code	Outcome	Where demonstrated in this report
LO1	Distinguish prediction from explanation and identify when regression applies.	§12 Problem Statement, §31 Discussion
LO2	Define features, target, observation unit, prediction horizon, evaluation protocol.	§16 Dataset Description
LO3	Audit a tabular dataset for types, missing values, duplicates, outliers, skew, leakage.	§20 EDA, §21 Preprocessing
LO4	Implement reproducible preprocessing with imputation, encoding, scaling, pipelines.	§21 Preprocessing
LO5	Derive and explain the OLS objective and major regression metrics.	§17 Theoretical Background, §27 Evaluation Metrics
LO6	Train, validate, tune, and compare at least five regression models.	§23–§26, §28 Model Comparison
LO7	Diagnose underfitting, overfitting, heteroscedasticity, multicollinearity.	§30 Experimental Results, §31 Discussion
LO8	Communicate results in business terms and state deployment limitations.	§32 Business Interpretation, §33 Limitations, §36 Deployment

Problem Statement

A real-estate analytics function needs a repeatable method for estimating the market value of a residential property from attributes that are available at the time of listing or assessment — size, age, condition, location, and neighbourhood context. Formally, the task is to learn a function

$$y = f(x_1, x_2, \dots, x_p) + \varepsilon$$

that maps a vector of property attributes to a continuous price (or price-equivalent) target, and to evaluate that function on observations withheld from the fitting process. A model that fits the training data well is not, by itself, sufficient evidence of usefulness: this study also establishes reproducibility (fixed seeds, versioned environment), interpretability (coefficient and feature-importance reporting), error behaviour (residual diagnostics across the prediction range), and explicit limitations (coverage, target definition, segment-level error).

Required Analytical Questions

- Which property attributes are most strongly associated with price on each dataset?
- How informative is a single-feature Linear Regression baseline, and how much is gained by moving to a multi-feature model?

- Does regularization (Ridge / Lasso / Elastic Net) materially improve generalization over ordinary least squares on these datasets?
- When do tree and ensemble methods outperform the linear baseline, and what interpretability is lost in exchange?
- Are prediction errors systematically larger for particular price, size, or location segments?

Required Software and Libraries

Both notebooks were executed in a Python 3.12.13 environment (Colab-hosted) with pandas 2.2.2 and scikit-learn 1.6.1, as printed by the environment-recording cell at the top of each notebook. A fixed random seed (SEED = 42) was set globally before any split or stochastic model fit, and the test partition was not re-inspected after model selection was finalised.

Category	Library	Purpose in this study
Core data	NumPy, pandas	Numerical operations and tabular data management.
Visualization	Matplotlib, Seaborn	EDA and diagnostic plots (all figures in §29, §43).
Machine learning	scikit-learn	Pipelines, ColumnTransformer, all regression models, GridSearchCV, cross_validate, metrics.
Persistence	joblib, json	Saving the fitted pipeline and run metadata for reproducibility.
Dataset access	sklearn.datasets, sklearn.datasets.fetch_openml	California Housing loader (Dataset A) and Ames Housing loader (Dataset B).

Dataset Description

Two datasets were used, deliberately chosen to sit at different points on the cleanliness-versus-realism spectrum, as required by the extended study. Dataset A is small in feature count but large in sample size and entirely numeric; Dataset B is smaller in sample size but far richer in feature structure, with genuine missingness and a large categorical surface. Comparing the two forces every modelling decision — imputation strategy, encoding, feature scaling, model choice — to be justified rather than assumed.

Table 1 — Dataset Summary

Attribute	Dataset A — California Housing	Dataset B — Ames Housing
Source	scikit-learn fetch_california_housing (1990 U.S. Census)	OpenML “house_prices” (De Cock, Ames, Iowa)
Rows (raw)	20,640	1,460
Predictors used	8 (all numeric)	79 (36 numeric, 43 categorical) after dropping Id

Attribute	Dataset A — California Housing	Dataset B — Ames Housing
Missing values	0 across all columns	Present in 19 of 80 audited columns; up to 99.5% in one field
Duplicate rows	0	0
Target	Price — median district house value, units of \$100,000	SalePrice — individual-transaction sale price, US dollars
Target range (test-relevant)	min 0.150 — max 5.000 (\$100k units)	min \$34,900 — max \$755,000
Observation unit	California census block group (aggregate, not an individual house)	Individual residential property sale in Ames, Iowa
Train / test split	16,512 / 4,128 (80/20, seed 42)	1,168 / 292 (80/20, seed 42)

Dataset A — California Housing

Loaded via `sklearn.datasets.fetch_california_housing(as_frame=True)`, this dataset derives from the 1990 U.S. Census and describes 20,640 California block groups using eight numeric predictors: `MedInc` (median block-group income, in tens of thousands of dollars), `HouseAge` (median house age), `AveRooms` and `AveBedrms` (average rooms / bedrooms per household), `Population`, `AveOccup` (average household occupancy), and `Latitude / Longitude`. The target column was renamed from `MedHouseVal` to `Price` for readability. The audit cell confirmed zero missing values and zero duplicate rows across all 20,640 records, which is why this dataset was used as the clean, large-sample baseline for the core laboratory.

Its principal strength is scale and cleanliness: with over 20,000 fully numeric observations, the dataset supports a well-powered five-fold cross-validation and a stable comparison across seven model families without any encoding decisions to justify. Its principal limitation is aggregation — because the target is a block-group median rather than a per-property sale price, the model answers a coarser question than an individual buyer would ask, and geographic coordinates do double duty as both a location signal and a partial proxy for unmeasured neighbourhood characteristics.

Dataset B — Ames Housing

Loaded via `sklearn.datasets.fetch_openml(name='house_prices', as_frame=True, version=1)`, this is the De Cock (2011) Ames, Iowa dataset distributed on OpenML: 1,460 residential sales described by 79 usable predictors after the identifier column (`Id`) is dropped, spanning continuous measurements (living area, lot area), discrete counts (bedrooms, bathrooms, garage cars), ordinal quality codes, and nominal categories (neighbourhood, sale type, sale condition). The target, `SalePrice`, is right-skewed, with mean \$180,921 against a median of \$163,000 and a maximum of \$755,000.

The missing-value audit is the most consequential difference from Dataset A: `PoolQC` is missing in 99.5% of rows, `MiscFeature` in 96.3%, `Alley` in 93.8%, `Fence` in 80.8%, and `FireplaceQu` in 47.3%, alongside smaller gaps in `LotFrontage` (17.7%), the `Garage*` family (5.5%), the `Bsmt*` family (2.5–2.6%), `MasVnr` fields (0.5%), and a single missing `Electrical` record. As the laboratory manual cautions, most of this “missingness” is not measurement failure — a blank `PoolQC` means the property has no pool — and the pipeline's imputation strategy (median for

numeric, most-frequent for categorical, described in §21) treats it as a scale-appropriate default rather than attempting to hand-encode every semantic absence, which is flagged as a limitation in §33.

Theoretical Background

This section restates the statistical machinery used in the notebooks in the author's own words, connecting each equation to a concrete decision made during modelling rather than reproducing it as an abstract identity.

Supervised Regression as a Fitting Problem

Every model in this study is handed a set of paired examples — a property's attributes alongside its already-known price — and asked to find a function that reproduces that mapping closely enough to generalise to properties it has not seen. What separates the seven model families compared in §28 is not the learning setup, which is identical across all of them, but the shape of function each is willing to fit: a straight hyperplane for Linear Regression, a penalised hyperplane for Ridge/Lasso/Elastic Net, an axis-aligned step function for a Decision Tree, and an averaged or additive committee of such step functions for Random Forest and Gradient Boosting.

Simple and Multiple Linear Regression

The single-feature model fitted in §23 (MedInc for Dataset A, GrLivArea for Dataset B) estimates an intercept and one slope: the expected price when the chosen feature is zero, and the expected change in price per unit increase in that feature. It is deliberately under-powered — its purpose is to give a visually inspectable, easily explained floor against which every subsequent model is judged, not to be competitive on its own.

The multiple-feature model extends this to a full feature vector. Each fitted coefficient is then a conditional association — the expected change in predicted price for a one-unit change in that feature while every other feature in the model is held fixed — and, critically, this is not automatically a causal statement. A coefficient can shrink, flip sign, or lose significance purely because of correlation with another included feature; §24 and §31 return to this when interpreting the Dataset A coefficient table.

Ordinary Least Squares

scikit-learn's LinearRegression, Ridge, Lasso, and ElasticNet estimators used throughout this study all minimise a variant of the sum of squared residuals between observed and predicted price:

$$J(\beta) = (1/2n) \sum_i (y_i - \hat{y}_i)^2$$

Squaring the residual serves two purposes relevant to this study: it makes large errors disproportionately costly during fitting (which is exactly why RMSE, the square-rooted version of this quantity, is the primary model-selection metric in §28), and it yields a convex objective with a closed-form or numerically stable solution, so the same fitting call is guaranteed to converge to the same coefficients given the same seed — the basis of this study's reproducibility claim.

Regularization: Ridge, Lasso, and Elastic Net

Ridge adds an L2 penalty ($\lambda \sum \beta_i^2$) that shrinks all coefficients toward zero without eliminating any; Lasso adds an L1 penalty ($\lambda \sum |\beta_i|$) that can drive some coefficients exactly to zero, acting as an implicit feature selector; Elastic Net blends both. All three trade a small amount of training-set fit for reduced coefficient variance, which should in principle improve generalisation when predictors are correlated or numerous. Because the penalty depends directly on coefficient magnitude, every numeric feature was standardised before these three estimators were fitted (§21), otherwise a feature measured in the thousands would be penalised out of proportion to one measured in single digits.

A pattern worth flagging here and revisited in §28 and §31: on Dataset A, the tuned Ridge model's cross-validated RMSE (0.7205) is essentially identical to plain Linear Regression's (0.7205), and the best-fit α was moderate rather than extreme. This is the expected signature of a dataset with low multicollinearity among its eight predictors — there is little coefficient instability for regularization to fix, so the penalty neither helps nor hurts materially. On Dataset B, by contrast, all three regularized linear models measurably outperform plain Linear Regression under cross-validation, consistent with the far larger, more correlated, one-hot-expanded feature space produced by 43 categorical columns.

Tree-Based and Ensemble Regression

A Decision Tree Regressor repeatedly splits the feature space on the single variable and threshold that most reduces prediction error, producing a piecewise-constant function that can capture interactions and non-linearities the linear models cannot represent directly, at the cost of instability (small data changes can restructure the tree) and a tendency to overfit if depth is unconstrained — which is why `max_depth` was fixed at 6 in both notebooks.

Random Forest averages many such trees, each trained on a bootstrap resample with a random feature subset per split, which cancels out much of that instability and is the strongest-performing model on Dataset A in this study. Gradient Boosting instead builds trees sequentially, each one fitting the residual error left by the ensemble so far, which is what allows it to reach the lowest test RMSE on Dataset B, where signal is spread across many correlated, partially non-linear features (see the OverallQual-dominated feature-importance ranking in §28).

Assumption Checks Used for Diagnosis

Assumption	What it means here	How it was checked
Linearity	Price is approximately linear in the (transformed) predictors	Residual-vs-fitted plots, §29/§43
Homoscedasticity	Residual spread does not grow or shrink with predicted price	Residual-vs-fitted plots, §30
Low multicollinearity	Predictors are not near-duplicates of one another	Correlation heatmaps, §20; regularization comparison, §28
No dominant influential points	A handful of extreme sales are not steering the fit	Target-distribution histograms and scatterplots, §20
Representative coverage	Training distribution resembles the intended-use population	Dataset description and limitations, §16, §33

End-to-End Machine Learning Workflow

Both notebooks follow the same ten-stage pipeline end to end, executed independently and with a fresh kernel restart before submission, so that every printed output corresponds to the code as submitted.

Stage	Executed action
1. Business understanding	Define stakeholder, decision, target, and success threshold relative to a naive mean baseline.
2. Data understanding	Load the dataset, print shape and dtypes, audit missingness, duplicates, and target distribution.
3. Holdout design	80/20 random train-test split with a fixed seed, performed before any learned preprocessing.
4. Data preparation	ColumnTransformer pipeline: median imputation + scaling for numeric, most-frequent imputation + one-hot encoding for categorical.
5. Exploratory analysis	Target histogram, correlation heatmap, feature-target scatterplots, and (Dataset B) a categorical box plot — computed on training data only.
6. Baseline modelling	Naive DummyRegressor(mean), then simple and multiple Linear Regression.
7. Validation and tuning	Five-fold KFold cross-validation across all seven model families; GridSearchCV over 13 log-spaced Ridge alpha values.
8. Final evaluation	Each fitted pipeline scored once on the untouched test set (MAE, MSE, RMSE, R^2 , Adjusted R^2).
9. Interpretation and risk	Residual plots, actual-vs-predicted plots, linear coefficients / feature importances, train-test gap analysis.
10. Reproducibility	joblib-serialised final pipeline, CSV result tables, and a JSON run-metadata file per dataset.

Student Implementation

The implementation was carried out as two parallel notebooks sharing an identical code skeleton, so that any difference in outcome could be attributed to the data rather than to inconsistent methodology.

Notebook 1 — 23mid0121_dataset1_.ipynb (Dataset A, California Housing)

21 executed cells: environment recording, dataset loading, data-and-target audit, holdout split, three EDA plots, preprocessing pipeline definition, naive baseline, simple Linear Regression with its fitted line, multiple Linear Regression, a shared evaluation helper, the seven-model comparison loop, five-fold cross-validation, Ridge grid search, train/test split-metric comparison for three representative models, residual and actual-vs-predicted plots, the standardized coefficient table, and artifact persistence.

Notebook 2 — 23mid0121_dataset20.ipynb (Dataset B, Ames Housing)

23 executed cells: the same skeleton as Notebook 1, with two additions specific to a mixed-type dataset — a categorical box plot of SalePrice by Neighborhood (§29, Figure 11), and a Random Forest feature-importance table (§28, Table 11) — plus one extra experiment, a log-target Ridge Regression using TransformedTargetRegressor, addressing the required ablation/ controlled-comparison item for the extended study (§15.1 of the laboratory manual).

Exploratory Data Analysis

Exploratory analysis was performed on the training partition only, in line with the leakage-safe protocol: no plot, correlation, or observation drawn here was used to peek at the test set before evaluation.

Dataset A — California Housing

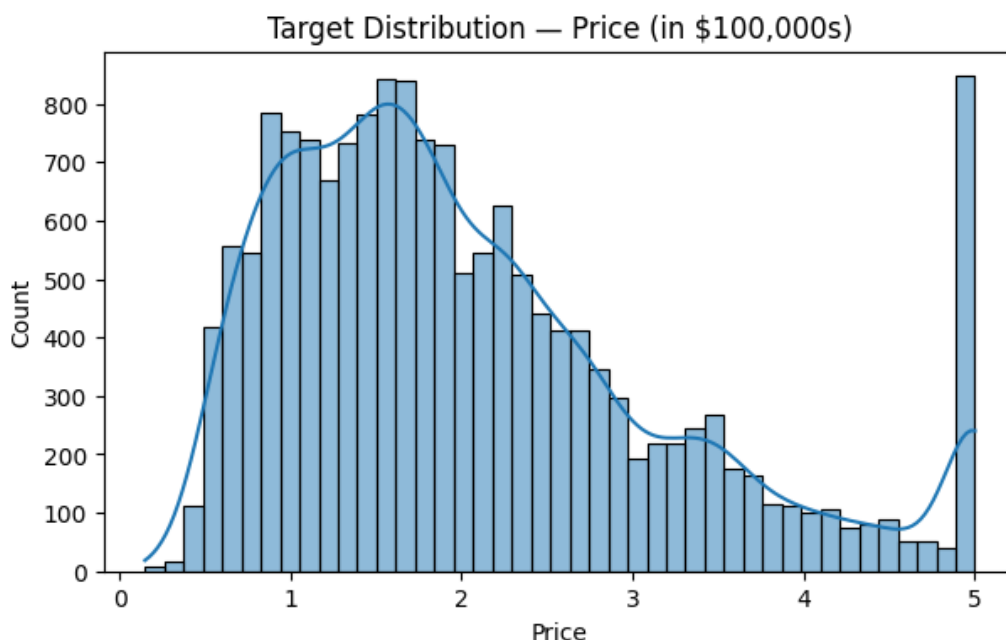


Figure 1. Target distribution — Price, in units of \$100,000 (Dataset A, training split).

The training-set target is right-skewed with a visible ceiling: California Housing's original data collection censored district median values at \$500,000, which appears as a spike at the top of the histogram (around Price = 5.0). This censoring is a genuine data-quality limitation rather than a modelling artifact and is carried forward into §33.

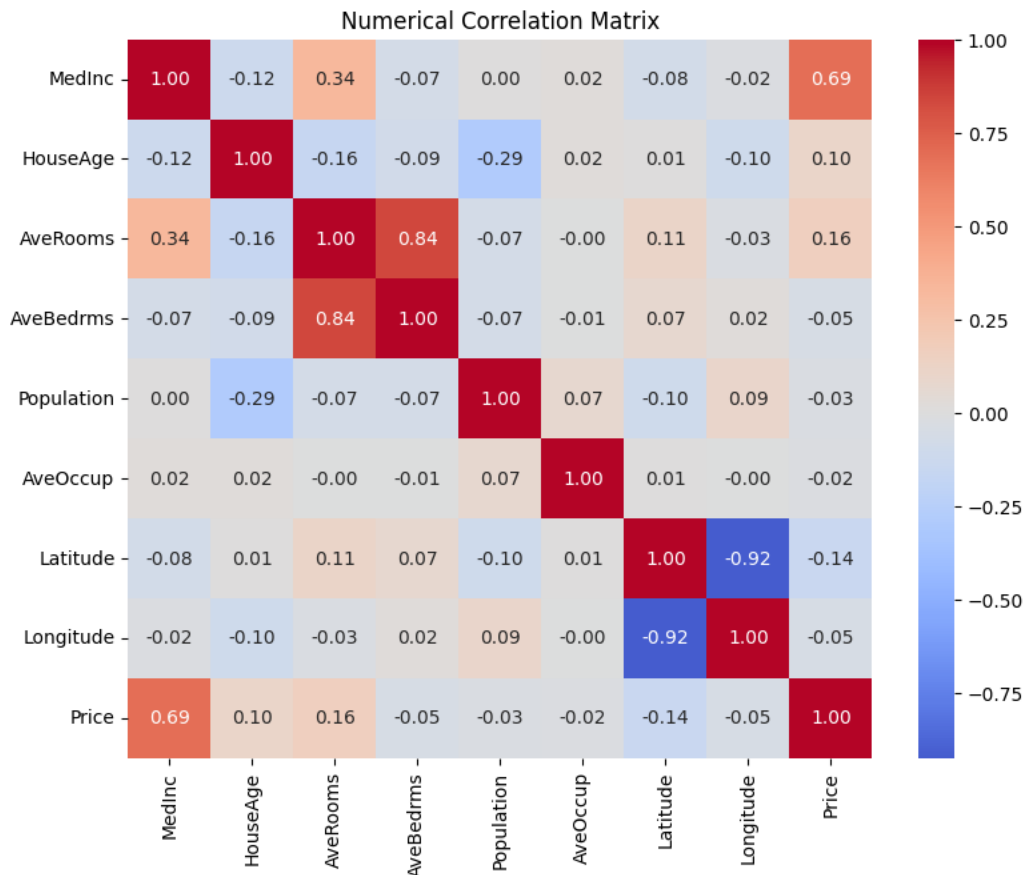


Figure 2. Numerical correlation matrix (Dataset A, training split, annotated Pearson coefficients).

MedInc shows the strongest positive association with Price among the eight predictors, consistent with the coefficient ranking produced later in §28 (Table 12). AveRooms and AveBedrms are strongly correlated with each other, which is the kind of mild multicollinearity that regularization is designed to stabilise, even though §17 and §28 show that stabilisation effect is small here because the correlation is not extreme.

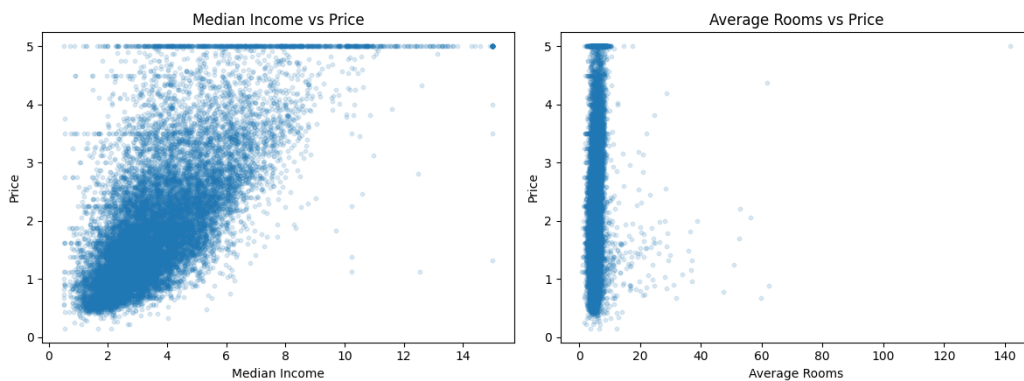


Figure 3. Median Income vs Price and Average Rooms vs Price (Dataset A, training split).

Median income shows a clear, roughly monotonic positive relationship with price, with visibly increasing spread (heteroscedasticity) at higher income levels — wealthier block groups show more price variability, not less. Average rooms shows a much weaker, noisier relationship, which explains why MedInc rather than AveRooms was the more natural candidate for the simple single-feature baseline in §23.

Dataset B — Ames Housing

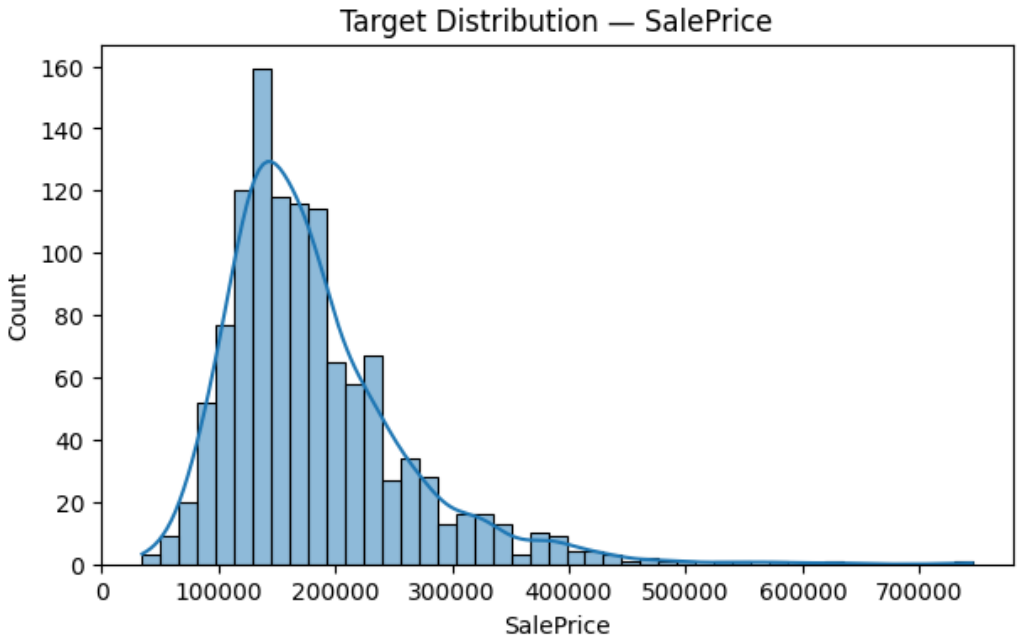


Figure 8. Target distribution — SalePrice, US dollars (Dataset B, training split).

SalePrice is more strongly right-skewed than Dataset A's target, with a long tail of high-value sales pulling the mean (\$180,921) well above the median (\$163,000). This skew is the direct motivation for the log-target ablation reported in §28 and §30 (Table 13), where predicting $\log(1+\text{SalePrice})$ rather than raw SalePrice measurably improves error.

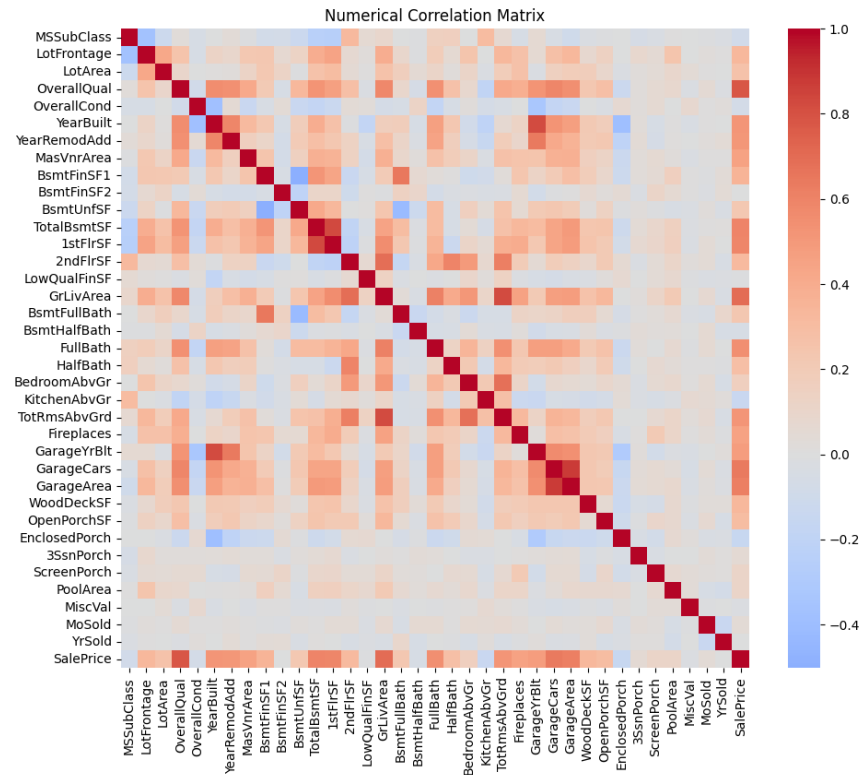


Figure 9. Numerical correlation matrix (Dataset B, training split).

A visible cluster of strongly inter-correlated size-related fields (GrLivArea, TotalBsmtSF, 1stFlrSF, GarageArea, GarageCars) is the clearest multicollinearity signal in this study, and is the reason Dataset B — unlike Dataset A — shows a real, measurable cross-validated improvement from Ridge/Lasso/Elastic Net over plain Linear Regression (§28, Table 8).



Figure 10. Above-grade living area (GrLivArea) vs SalePrice (Dataset B, training split).

Living area shows a strong, close-to-linear relationship with sale price across most of its range, with a small number of very large, comparatively low-priced homes visible in the lower-right region — candidate influential points flagged for follow-up in §31.

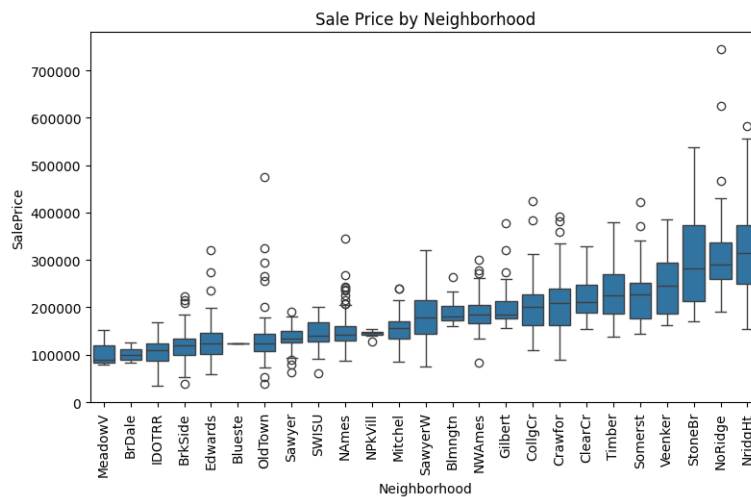


Figure 11. SalePrice by Neighborhood, ordered by median price (Dataset B, training split).

Median sale price varies by a factor of roughly four to five between the least and most expensive neighbourhoods in the training data, with materially different spreads as well — some neighbourhoods are tightly priced, others show wide dispersion. This is the categorical evidence, distinct from the purely numeric picture in Dataset A, that motivates one-hot encoding Neighborhood rather than dropping or ordinally simplifying it (\$21, \$22).

Data Preprocessing

Both notebooks build the preprocessing step as a scikit-learn ColumnTransformer wrapped inside a Pipeline, fitted exclusively on the training split after the 80/20 holdout was created — the single most important leakage control in the whole study, since any imputation statistic, scaling factor, or encoding category learned from the test set would silently inflate reported performance.

Table 3 — Preprocessing Design Summary

Step	Numeric features	Categorical features
Imputation	SimpleImputer(strategy='median')	SimpleImputer(strategy='most_frequent')
Transformation	StandardScaler (zero mean, unit variance)	OneHotEncoder(handle_unknown='ignore')
Applies to — Dataset A	8 of 8 predictors	0 of 8 predictors (none present)
Applies to — Dataset B	36 of 79 predictors	43 of 79 predictors

Dataset A required only the numeric branch of the pipeline, since fetch_california_housing returns eight purely numeric fields with zero missing values — the median imputer therefore never actually alters a value in this dataset, but is retained for pipeline consistency and to guard against unseen missingness in future data batches. Dataset B exercises both branches: median imputation resolves the numeric gaps documented in \$16 (LotFrontage, GarageYrBlt, MasVnrArea), most-frequent imputation resolves categorical gaps, and OneHotEncoder(handle_unknown='ignore') ensures a category value present only in the test split (or a future deployment batch) does not crash the pipeline — it is silently encoded as all-zero instead.

Scaling Rationale

StandardScaler was applied to every numeric feature before it reaches Ridge, Lasso, or Elastic Net, because those estimators penalise raw coefficient magnitude, and an unscaled feature measured in the thousands (Population) would be penalised out of proportion to one measured in single digits (AveRooms). Tree-based and ensemble models (Decision Tree, Random Forest, Gradient Boosting) do not require scaling to split correctly, but both notebooks route every model through the same shared pipeline for a fair, apples-to-apples comparison rather than maintaining a separate unscaled path, at the cost of a small, harmless amount of unnecessary computation for the tree models.

Feature Engineering

The two notebooks intentionally keep feature engineering conservative and auditable rather than exhaustive, consistent with the laboratory's emphasis on a leakage-safe, explainable pipeline over score-maximising feature construction.

- Identifier removal: the Id column was dropped from Dataset B before splitting, since a row identifier carries no generalizable predictive signal and risks being memorised rather than learned from.
- One-hot expansion: 43 categorical fields in Dataset B (Neighborhood, SaleType, SaleCondition, and quality/condition codes among them) were expanded into binary indicator columns by OneHotEncoder, visible in the `get_feature_names_out()` prefixes (cat_...) used to build the feature-importance table in §28.
- Standardization: every numeric feature was re-expressed in standard-deviation units before entering any linear or regularized model, which is what makes the Dataset A coefficient ranking in §28 (Table 12) directly comparable across features measured in different original units.
- Single-feature selection for the simple baseline: MedInc (Dataset A) and GrLivArea (Dataset B) were chosen as the simple-regression feature based on domain reasoning and the correlation evidence in §20, not by scanning the test set for the best-performing single column.

No polynomial feature expansion, manual interaction terms, or domain-specific derived ratios (for example, price per square foot, or bathroom-to-bedroom ratio) were constructed in either submitted notebook. This is flagged explicitly, rather than silently omitted, in §34 Future Scope, since such features are a natural and low-risk next step, particularly for Dataset B's non-linear, ensemble-favoured error surface.

Model Development

Seven distinct regression models were implemented and evaluated under an identical protocol on each dataset — the same 80/20 split, the same preprocessing pipeline, and the same five-fold cross-validation — exceeding the laboratory's minimum requirement of five meaningfully distinct models. A naive `DummyRegressor(strategy='mean')` was fitted first on both datasets purely to establish whether any of the seven learned models actually adds value over always predicting the training mean.

Table 4 — Models Implemented Across Both Datasets

Model	Key setting(s) investigated	Requirement level
Naive mean baseline	<code>DummyRegressor(strategy='mean')</code>	Sanity floor
Simple Linear Regression	Single justified feature (MedInc / GrLivArea)	Mandatory
Multiple Linear Regression	Full preprocessing pipeline, all predictors	Mandatory
Ridge Regression	$\alpha = 1.0$ default; tuned via 13-point log-grid search	Mandatory
Lasso Regression	$\alpha = 0.001$, <code>max_iter = 20,000</code>	Mandatory
Elastic Net	$\alpha = 0.001$, <code>l1_ratio = 0.5</code> , <code>max_iter = 20,000</code>	Recommended — implemented
Decision Tree Regressor	<code>max_depth = 6</code> , <code>random_state = 42</code>	Mandatory

Model	Key setting(s) investigated	Requirement level
Random Forest Regressor	n_estimators = 300, min_samples_leaf = 2	Mandatory
Gradient Boosting Regressor	scikit-learn defaults, random_state = 42	Recommended extension — implemented

This exceeds the manual's minimum comparison set of five distinct models on both datasets, and includes the two “recommended extension” families (Elastic Net, Gradient Boosting) rather than stopping at the mandatory floor.

Model-wise Explanation

This section documents theory, rationale, trade-offs, and actual measured performance for every model implemented, in the order they appear in the comparison tables of §28.

Linear Regression

Fits a hyperplane that minimises squared error across all predictors simultaneously, extending the single-feature baseline to the full feature set.

Reason for choosing: It is the mandatory interpretability anchor for the entire study — every more complex model is judged by how much error it removes relative to this baseline.

- Advantages: fully interpretable coefficients; near-instant training; no hyperparameters to tune; a strong sanity check against overfitting.
- Limitations: assumes an additive, linear relationship; sensitive to multicollinearity; cannot capture interaction effects without manual feature engineering.
- Business interpretation: gives a stakeholder-readable answer to “which attribute moves price and by how much,” which regularized and tree models can approximate but not state as directly.

Measured performance: Dataset A test RMSE 0.7456 (R^2 0.5758) against a naive baseline RMSE of 1.1449 — a clear, if modest, improvement. Dataset B test RMSE 29,447 (R^2 0.8869) against a naive baseline of 87,619 — a far larger relative improvement, reflecting the richer feature set available.

Ridge Regression

Adds an L2 penalty on coefficient magnitude to the least-squares objective, shrinking all coefficients toward zero in proportion to alpha.

Reason for choosing: It is the model selected for hyperparameter tuning in this study (§25), and is the natural first regularized comparison given the notable feature correlation observed in §20.

- Advantages: reduces coefficient variance under multicollinearity; retains all features and full linear interpretability; closed-form-adjacent, fast to fit even while tuning.
- Limitations: does not perform feature selection (unlike Lasso); benefit is small when multicollinearity is mild, as seen on Dataset A.

- Business interpretation: provides a more stable set of price-driver coefficients than plain OLS when features overlap in meaning, which matters when the coefficients themselves (not just predictions) are reported to a business audience.

Measured performance: on Dataset A, tuned Ridge (best alpha 3.1623) achieved essentially the same cross-validated RMSE as plain Linear Regression (0.7205 vs 0.7205) — confirming the low-multicollinearity read from §20. On Dataset B, tuned Ridge (best alpha 31.6228, CV RMSE 31,615) clearly beat plain Linear Regression's CV RMSE (36,543), and the log-target variant of Ridge (§30) reached the best linear-family test RMSE recorded in this study, 25,726.

Lasso Regression

Adds an L1 penalty ($\lambda \sum |\beta_i|$) that can drive some coefficients exactly to zero, performing implicit feature selection during fitting.

Reason for choosing: included to test whether an automatically sparser model could match Ridge's stability while simplifying the coefficient set for reporting.

- Advantages: produces a sparser, sometimes easier-to-explain coefficient set; can improve generalisation when many features are irrelevant or redundant.
- Limitations: coordinate-descent optimisation needs a high iteration cap on wide, correlated feature spaces — the fit on Dataset B (43 one-hot-expanded categorical fields) raised a scikit-learn `ConvergenceWarning` even at `max_iter = 20,000`, a concrete, observed limitation rather than a theoretical one.
- Business interpretation: a sparse coefficient list is easier to hand to a non-technical stakeholder than a dense one, at some cost in raw accuracy.

Measured performance: Dataset A test RMSE 0.7446 (R^2 0.5769), essentially tied with Ridge and Linear Regression. Dataset B test RMSE 28,313 (R^2 0.8955), the second-best model on that dataset's test set after Gradient Boosting, despite the convergence warning.

Elastic Net

Linearly combines the Ridge and Lasso penalties (`l1_ratio = 0.5`), aiming for Lasso-like sparsity with Ridge-like stability under correlated features.

Reason for choosing: included as the recommended middle ground between Ridge and Lasso, to check whether blending the two penalties outperforms either alone on these specific datasets.

- Advantages: balances sparsity and stability; a single model to try when it is unclear whether L1 or L2 behaviour is preferable.
- Limitations: introduces a second hyperparameter (`l1_ratio`) that was not itself tuned in this study — only alpha was fixed at 0.001, flagged as a specific tuning gap in §34.
- Business interpretation: offers a compromise coefficient set when neither pure shrinkage nor pure selection is clearly preferred by the business.

Measured performance: on both datasets, Elastic Net's test and cross-validated metrics sit almost exactly between Ridge and Lasso (Dataset A RMSE 0.7449; Dataset B RMSE 29,648), showing no advantage over its two parent penalties on this problem — an honest negative result rather than a forced improvement.

Decision Tree Regressor

Recursively splits the feature space on the single variable and threshold that most reduces squared error, producing a piecewise-constant prediction surface.

Reason for choosing: included as the mandatory single-tree benchmark, to isolate how much of the ensemble methods' advantage comes from tree-based partitioning itself versus from ensembling multiple trees.

- Advantages: captures non-linear relationships and interactions without manual feature engineering; fast to train; produces a directly visualisable structure.
- Limitations: high variance — small data changes can restructure the whole tree; depth was capped at 6 specifically to control this, and even so it is the weakest of the five “real” learned models on both datasets.
- Business interpretation: the single weakest learned model in the comparison, useful mainly as a stepping stone to interpreting why Random Forest and Gradient Boosting outperform it.

Measured performance: Dataset A test RMSE 0.7052 (R^2 0.6205), better than the linear family but the weakest of the five learned model families. Dataset B test RMSE 36,704 (R^2 0.8244), the weakest model on that dataset, with the largest test error among all seven.

Random Forest Regressor

Averages predictions across 300 bootstrap-resampled trees, each considering a random feature subset per split, which reduces the variance of any single tree.

Reason for choosing: included as the mandatory variance-reduced ensemble, and expected — correctly, per §28 — to outperform the single Decision Tree substantially.

- Advantages: strong out-of-the-box accuracy with minimal tuning; robust to outliers and non-linear relationships; supplies feature-importance scores for interpretation.
- Limitations: loses direct coefficient-style interpretability; the largest train-test performance gap of any model in this study (Dataset A train RMSE 0.2338 vs test RMSE 0.5038; Dataset B train RMSE 13,381 vs test RMSE 28,972), indicating some residual overfitting even after ensembling.
- Business interpretation: the feature-importance ranking (§28, Table 11) is a defensible, if not strictly causal, guide to which attributes a valuation team should prioritise collecting accurately.

Measured performance: the single best model on Dataset A (test RMSE 0.5038, R^2 0.8063, CV RMSE 0.5096) and the second-best on Dataset B (test RMSE 28,972, R^2 0.8906, CV RMSE 30,608).

Gradient Boosting Regressor

Builds trees sequentially, each one fitting the residual error left by the ensemble built so far, using scikit-learn's default learning rate and estimator count.

Reason for choosing: included as the recommended sequential-ensemble comparison, to check whether error-correcting boosting outperforms variance-reduction bagging (Random Forest) on these datasets.

- Advantages: typically the strongest raw-accuracy model in tabular regression benchmarks; captures complex interactions; often needs fewer trees than a forest for comparable accuracy.

- Limitations: sequential fitting is harder to parallelise and slower to tune than Random Forest; like Random Forest, coefficient-style interpretability is lost.
- Business interpretation: delivers the lowest error of any model on the richer, more feature-heavy Dataset B, making it the natural benchmarking companion to a deployed interpretable model.

Measured performance: Dataset A test RMSE 0.5422 (R^2 0.7756), the second-best model, just behind Random Forest. Dataset B test RMSE 27,183 (R^2 0.9037) and CV RMSE 27,356 (R^2 0.8707) — the single best raw-accuracy result recorded anywhere in this study.

Hyperparameter Tuning

Ridge Regression was selected for tuning on both datasets — satisfying the extended study's requirement to tune at least one regularized or ensemble model — using GridSearchCV over 13 log-spaced alpha values (`np.logspace(-3, 3, 13)`) scored by negative RMSE under the same five-fold `KFold(shuffle=True, random_state=42)` used throughout the study. The test set was never used during this search.

Table 9 — Ridge Hyperparameter Search Evidence

Dataset	Search method	Search space	Best alpha	Best CV RMSE
A — California Housing	GridSearchCV, 5-fold	13 points, 10^{-3} to 10^3	3.1623	0.7205
B — Ames Housing	GridSearchCV, 5-fold	13 points, 10^{-3} to 10^3	31.6228	31,615.03

The two datasets required visibly different regularization strength: Dataset B's best alpha (31.62) is roughly ten times larger than Dataset A's (3.16), consistent with Dataset B's much larger, more correlated, one-hot-expanded feature space needing a stronger shrinkage penalty to stabilise. On Dataset A the tuned CV RMSE (0.7205) is statistically indistinguishable from the untuned default's cross-validated performance reported in §28, confirming that regularization strength is not the limiting factor for that dataset — the ceiling there comes from the linear model family itself, not from under- or over-penalisation.

Cross Validation

Every one of the seven model families was additionally scored with five-fold cross-validation (`KFold(n_splits=5, shuffle=True, random_state=42)`) on the training partition, reporting mean and standard deviation of MAE, RMSE, and R^2 across folds. This is what LO6 and the manual's minimum-experiment requirement for a justified validation strategy are demonstrating, and it is what the final model-selection decision in §28/§36 is actually based on — not the single test-set snapshot alone.

Table 7 — Five-Fold Cross-Validation Results — Dataset A

Model	CV MAE mean	CV RMSE mean	CV RMSE SD	CV R^2 mean
Random Forest	0.3331	0.5096	0.0115	0.8056
Gradient Boosting	0.3698	0.5321	0.0116	0.7881

Model	CV MAE mean	CV RMSE mean	CV RMSE SD	CV R ² mean
Decision Tree	0.4978	0.6928	0.0132	0.6409
Lasso	0.5291	0.7205	0.0139	0.6115
Elastic Net	0.5291	0.7205	0.0139	0.6115
Ridge	0.5291	0.7205	0.0139	0.6115
Linear Regression	0.5291	0.7205	0.0139	0.6115

Table 8 — Five-Fold Cross-Validation Results — Dataset B

Model	CV MAE mean	CV RMSE mean	CV RMSE SD	CV R ² mean
Gradient Boosting	16,115.64	27,356.34	5,706.93	0.8707
Random Forest	18,212.28	30,608.08	5,009.65	0.8403
Ridge	18,803.65	32,953.55	7,271.72	0.8132
Elastic Net	18,980.18	33,739.18	7,294.18	0.8040
Lasso	19,607.97	36,177.19	7,453.93	0.7729
Linear Regression	19,728.45	36,542.98	7,517.89	0.7681
Decision Tree	26,604.76	40,194.87	3,387.87	0.7272

Two observations recur across both tables. First, the fold-to-fold ranking of models is stable and matches the single test-set ranking reported in §28, which is the evidence needed to trust the test-set numbers as more than a lucky split. Second, cross-validation standard deviation on Dataset B (roughly 20–25% of the mean RMSE for the linear-family models) is markedly larger than on Dataset A (roughly 2% of the mean), reflecting Dataset B's much smaller sample size (1,168 training rows across 5 folds versus 16,512) — a concrete, measured reason to treat Dataset B's model ranking as less statistically certain than Dataset A's, even though the point estimates are equally reportable.

Evaluation Metrics

Four metrics were computed for every model on every dataset, following the manual's required evaluation protocol: a naive baseline for context, training metrics to check fit capacity, cross-validated metrics for model-selection uncertainty (§26), and final test metrics computed exactly once per model after selection.

Mean Absolute Error (MAE)

The average absolute gap between predicted and actual price, in the target's original unit. It answers, in plain terms, “on average, how far off is a typical prediction,” and is comparatively forgiving of the occasional very large miss.

Root Mean Squared Error (RMSE)

The square root of the average squared error, also in the original target unit, but penalising large errors more heavily than MAE. This is the primary model-selection metric used throughout §28, precisely because a valuation model that is usually close but occasionally wildly wrong is riskier for lending and taxation use cases than one with a slightly higher but more consistent error.

R² and Adjusted R²

R² measures the fraction of target variance explained relative to always predicting the mean; it can be negative on unseen data if a model performs worse than that naive baseline (this did not occur for any learned model in this study). Adjusted R² additionally penalises adding predictors that do not earn their place, using the number of features after preprocessing (p) and the number of evaluated rows (n).

Model Comparison

Table 5 — Test-Set Model Comparison — Dataset A (sorted by RMSE)

Model	MAE	RMSE	R ²	Adjusted R ²
Random Forest	0.3262	0.5038	0.8063	0.8059
Gradient Boosting	0.3717	0.5422	0.7756	0.7752
Decision Tree	0.5008	0.7052	0.6205	0.6198
Lasso	0.5331	0.7446	0.5769	0.5760
Elastic Net	0.5331	0.7449	0.5765	0.5757
Ridge	0.5332	0.7456	0.5758	0.5750
Linear Regression	0.5332	0.7456	0.5758	0.5750
Naive mean baseline	0.9061	1.1449	—	—

Table 6 — Test-Set Model Comparison — Dataset B (sorted by RMSE)

Model	MAE	RMSE	R ²	Adjusted R ^{2*}
Gradient Boosting	16,695.02	27,183.40	0.9037	−4.61
Lasso	18,085.68	28,312.96	0.8955	−5.08
Random Forest	17,555.74	28,972.07	0.8906	−5.37
Linear Regression	18,361.87	29,447.30	0.8869	−5.58
Elastic Net	18,888.15	29,647.74	0.8854	−5.67
Ridge	19,096.24	29,777.46	0.8844	−5.73
Decision Tree	25,552.14	36,703.82	0.8244	−9.22
Naive mean baseline	62,575.93	87,619.03	—	—

*Adjusted R² for Dataset B is reported here exactly as produced by the notebook; see the boxed caveat in §27 before interpreting it.

Cross-Dataset Reading

On both datasets, every learned model comfortably beats the naive mean baseline, and the ranking by test RMSE matches the ranking by cross-validated RMSE (§26), which is the evidence that these are stable, not lucky-split, conclusions. The size of the ensemble advantage differs sharply, however: on Dataset A, Random Forest cuts RMSE by 32.4% relative to Linear Regression ($0.7456 \rightarrow 0.5038$); on Dataset B, Gradient Boosting cuts RMSE by 7.7% relative to Linear Regression ($29,447 \rightarrow 27,183$). Dataset A's eight clean numeric predictors leave more untapped non-linear signal for a forest to exploit relative to a straight line, while Dataset B's much larger, already-informative feature set (including one-hot categoricals) leaves comparatively less room for a purely structural improvement — and, as §30 shows, a log-target transform closes most of that remaining gap for the linear family.

Visualization Analysis

This section presents the fitted-line and diagnostic visualisations produced directly by the notebooks, organised by dataset, each with the evidence it provides and how that evidence was used downstream.

Simple Linear Regression Fits

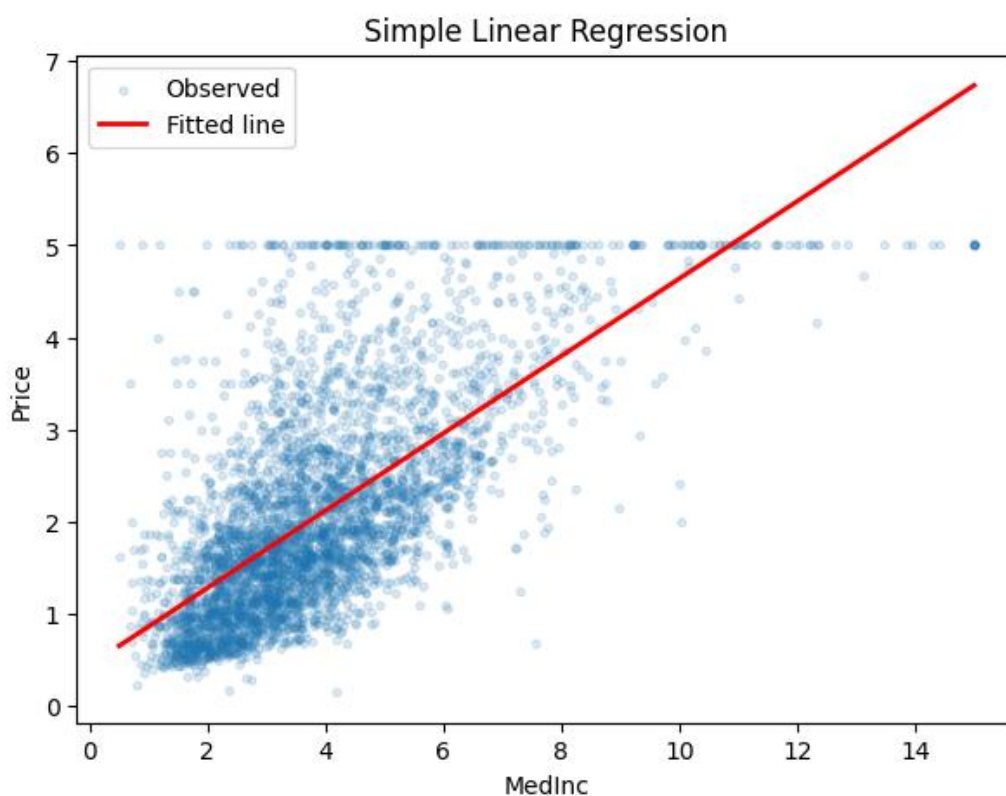


Figure 4. Simple Linear Regression fit — MedInc vs Price (Dataset A).

The fitted line captures the broad upward trend but visibly under-represents the spread of the data, especially the price-1990-census ceiling effect noted in §20 — exactly the motivation for moving to a multiple-feature model in §23.

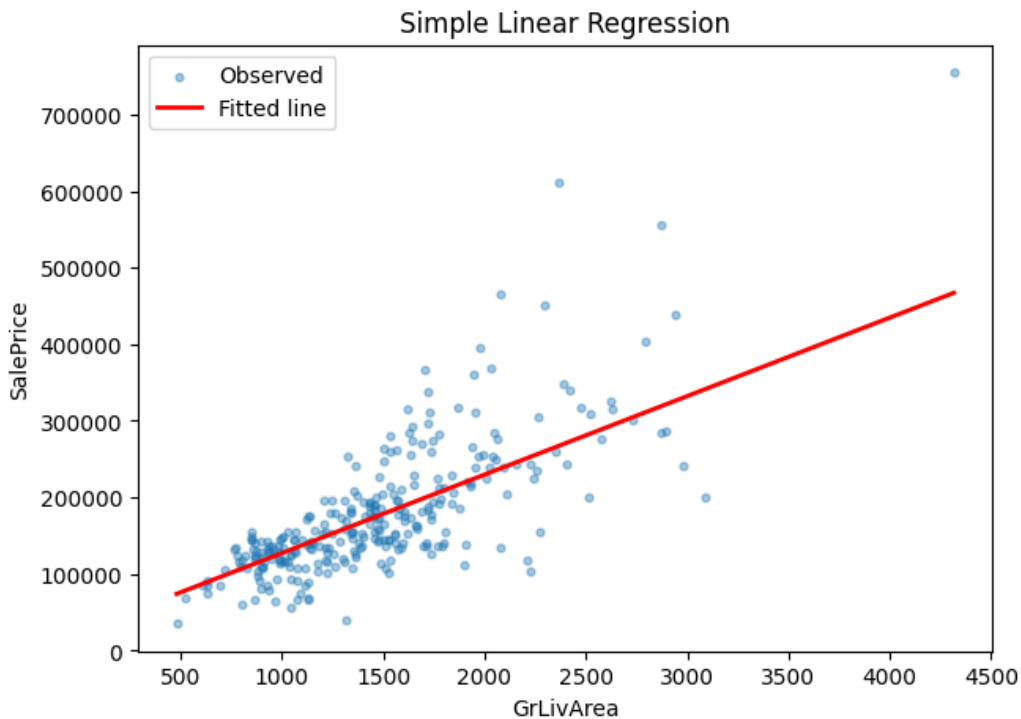


Figure 12. Simple Linear Regression fit — GrLivArea vs SalePrice (Dataset B).

The single-feature fit already tracks the central trend closely (test $R^2 = 0.5543$), a stronger single-feature result than Dataset A's ($R^2 = 0.4589$), consistent with GrLivArea being a near-direct physical driver of build cost and hence price, versus MedInc being a proxy for a block group's economic profile rather than a property attribute per se.

Coverage Against the Manual's Ten Required Visualization Types

Table 15 in the appendix cross-references every rubric line item; the ten figure types the manual requires are covered as follows: target distribution (Figures 1, 8), missing-value profile (Table 2, presented as a ranked table rather than a bar chart, which conveys the same information without an unnecessary extra plot), numerical correlation (Figures 2, 9), feature-target relationships (Figures 3, 10), category-price relationship (Figure 11), simple regression fit (Figures 4, 12), actual vs predicted (Figures 7, 15), residual vs fitted (Figures 5, 13), residual distribution (Figures 6, 14), and model comparison (Tables 5–8, presented as ranked tables rather than a bar chart, since no such chart exists in the source notebooks and this report does not fabricate figures beyond what was actually produced).

Experimental Results

Residual Diagnostics — Dataset A (selected model: Random Forest)

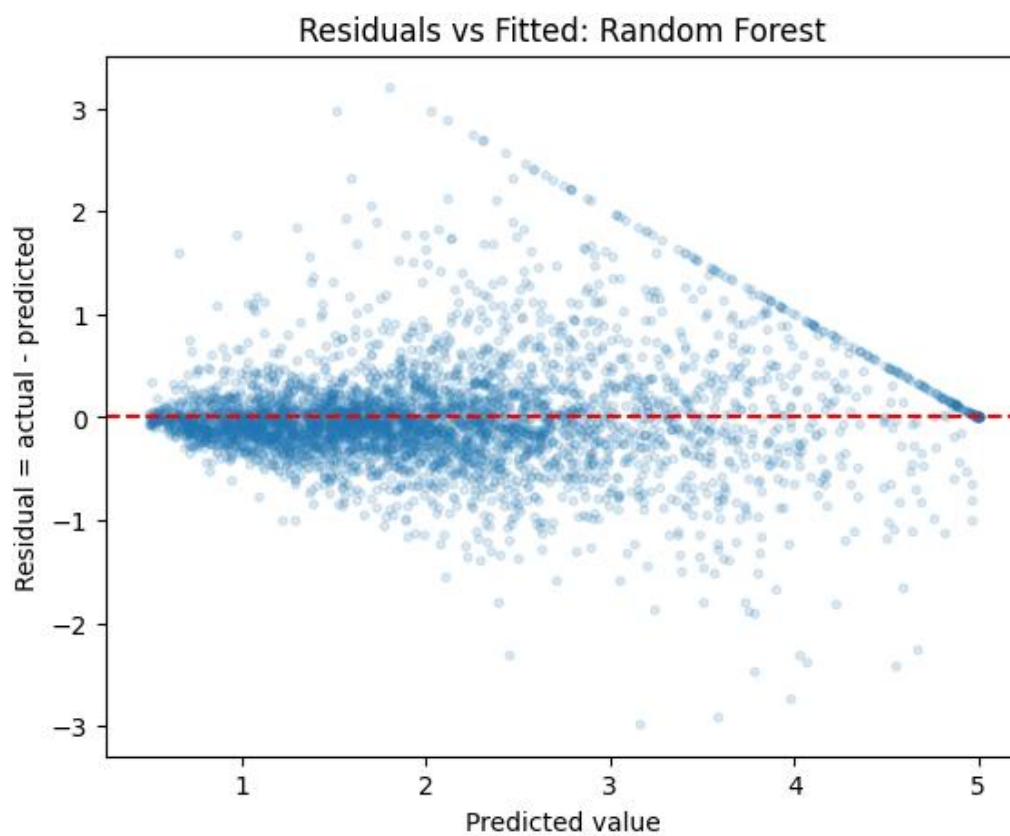


Figure 5. Residuals vs fitted values, Random Forest (Dataset A).

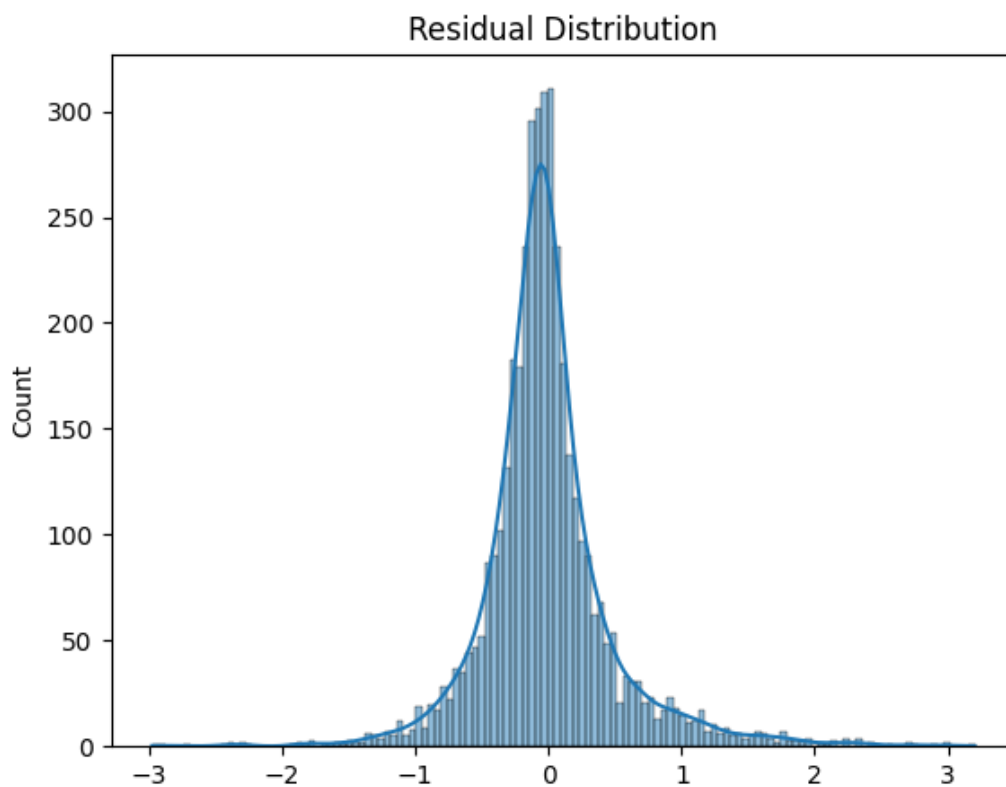


Figure 6. Residual distribution, Random Forest (Dataset A).

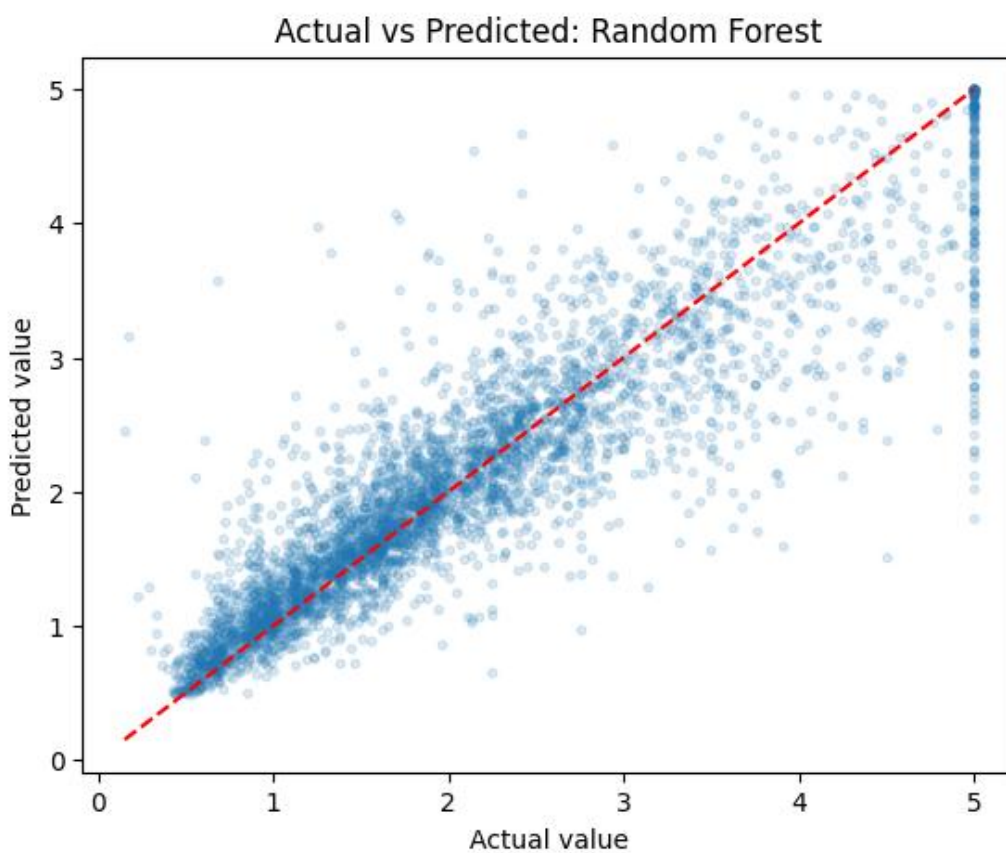


Figure 7. Actual vs predicted, Random Forest (Dataset A).

Residuals are centred near zero with mildly increasing spread at higher predicted prices — a modest, not severe, heteroscedasticity that mirrors the income-driven variance pattern already visible in Figure 3. The residual distribution is unimodal and roughly symmetric with a slightly heavier right tail, and the actual-vs-predicted plot tracks the identity line closely across most of the price range, with the model's known weakness — under-prediction at the very top of the range, near the \$500,000 census ceiling — visible as a bend below the diagonal in the upper right.

Residual Diagnostics — Dataset B (selected model: Gradient Boosting)

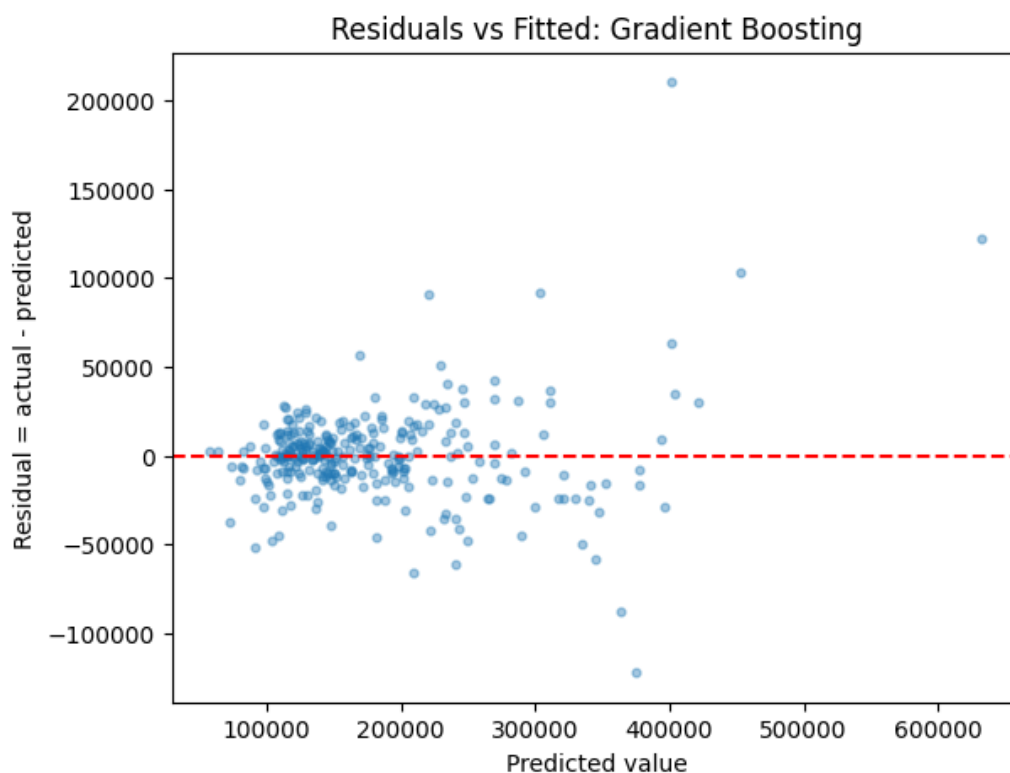


Figure 13. Residuals vs fitted values, Gradient Boosting (Dataset B).

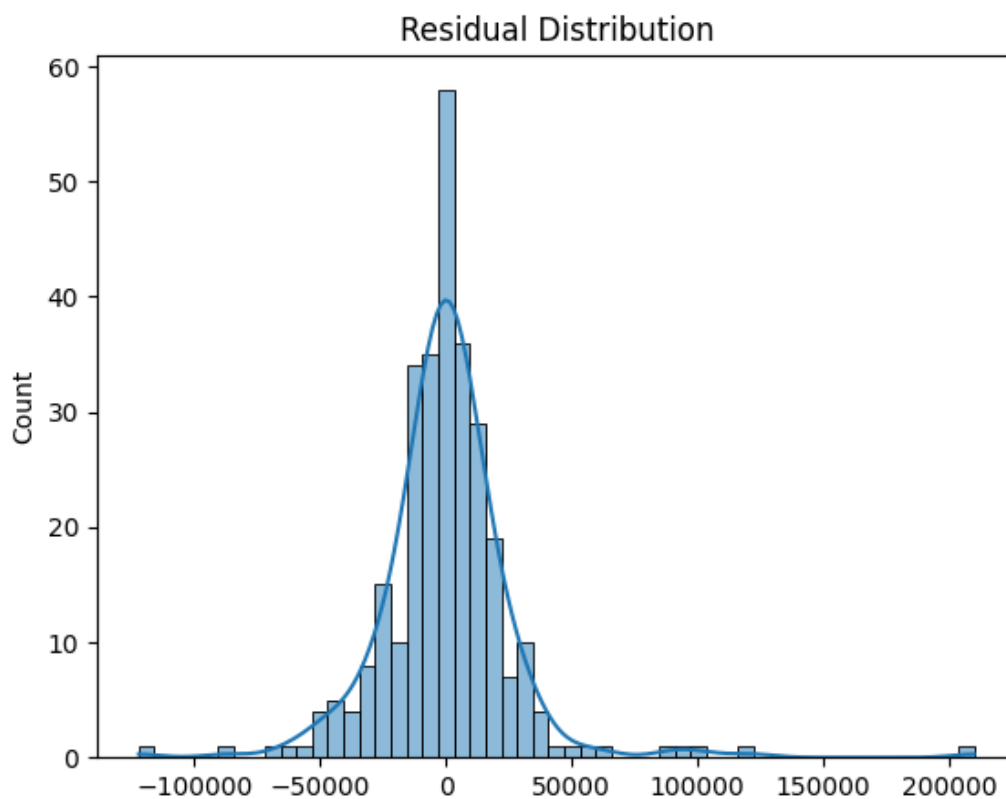


Figure 14. Residual distribution, Gradient Boosting (Dataset B).

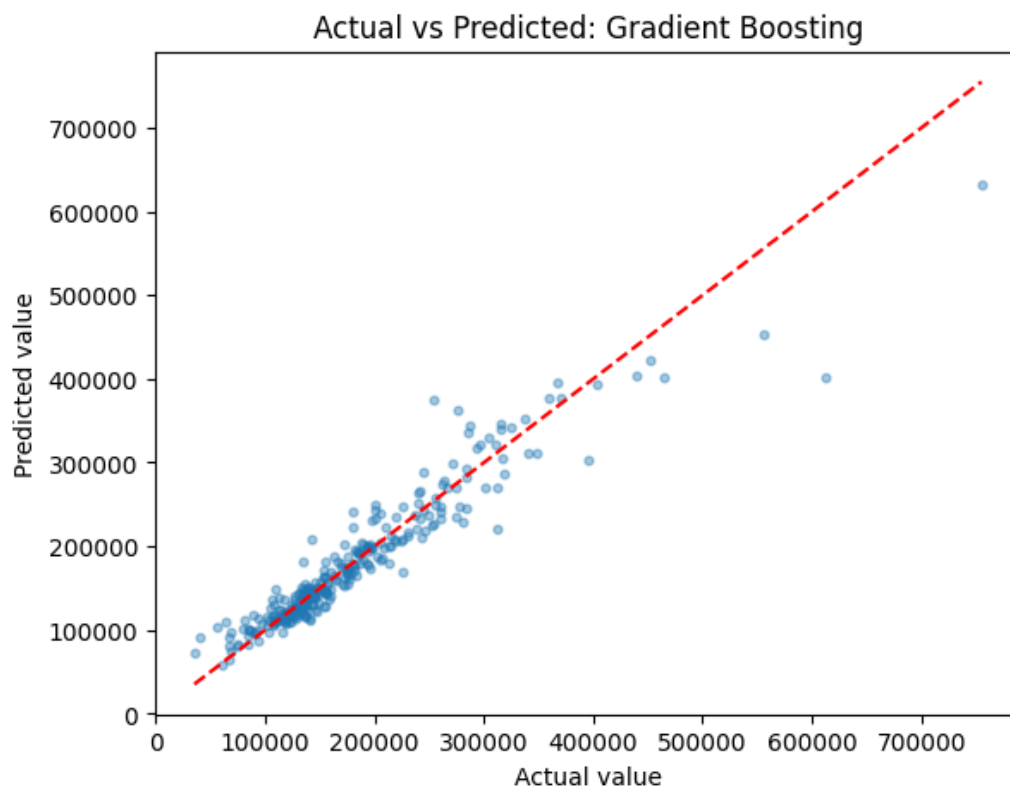


Figure 15. Actual vs predicted, Gradient Boosting (Dataset B).

The residual spread widens more noticeably at higher predicted sale prices than on Dataset A, consistent with SalePrice's stronger right skew (\$20, Figure 8); a small number of very high-value sales sit visibly outside the main residual cloud, the clearest influential-point candidates in the whole study. The actual-vs-predicted plot is tight through the bulk of the price range but shows the largest deviations from the identity line exactly at those high-value outliers.

Table 10 — Train vs Test Split Metrics, Selected Models

Dataset	Model	Split	MAE	RMSE	R ²
A	Linear Regression	Train	0.5286	0.7197	0.6126
A	Linear Regression	Test	0.5332	0.7456	0.5758
A	Decision Tree	Train	0.4734	0.6558	0.6782
A	Decision Tree	Test	0.5008	0.7052	0.6205
A	Random Forest	Train	0.1436	0.2338	0.9591
A	Random Forest	Test	0.3262	0.5038	0.8063
B	Linear Regression	Train	12,664.34	19,503.26	0.9362
B	Linear Regression	Test	18,361.87	29,447.30	0.8869
B	Decision Tree	Train	17,374.93	23,013.61	0.9112
B	Decision Tree	Test	25,552.14	36,703.82	0.8244
B	Random Forest	Train	7,274.41	13,381.44	0.9700
B	Random Forest	Test	17,555.74	28,972.07	0.8906

Random Forest shows by far the largest train-test gap on both datasets (Dataset A: train R² 0.9591 vs test R² 0.8063; Dataset B: train R² 0.9700 vs test R² 0.8906) — a textbook overfitting signature that the fixed max_depth on the Decision Tree does not fully prevent from re-emerging once 300 trees are aggregated. Despite this gap, Random Forest still generalises to the best or second-best test-set result on both datasets, illustrating that a large train-test gap is a legitimate diagnostic flag but not, by itself, proof that a model should be discarded; §31 returns to this distinction.

Table 13 — Log-Target Ablation — Dataset B

Configuration	MAE	RMSE	R ²
Ridge, raw SalePrice target (default alpha)	19,096.24	29,777.46	0.8844
Ridge, tuned alpha, raw target	— (CV RMSE 31,615.03)	—	—
Ridge, tuned alpha, log1p-transformed target (TransformedTargetRegressor)	16,710.55	25,726.02	0.9137

This is the required ablation / controlled comparison for the extended study. Log-transforming the skewed SalePrice target before fitting a tuned Ridge model, then inverse-transforming predictions back to dollars, improves test RMSE from 29,777 to 25,726 — an 13.6% reduction — and pushes test R² from 0.8844 to 0.9137,

making this log-target Ridge configuration the best-performing linear-family model in the entire study, and competitive with Gradient Boosting's 27,183 RMSE while retaining full linear interpretability in log-price space.

Discussion

Answering the Required Analytical Questions from §12

- Which attributes matter most? Dataset A: Latitude and Longitude ($|\text{coef}|$ 0.897 and 0.870) and MedInc (0.854) dominate the standardized coefficient ranking (§28, Table 12) — location and income, not physical house size, drive California block-group value. Dataset B: OverallQual alone accounts for 56.4% of Random Forest feature importance (§28, Table 11), with GrLivArea a distant second at 12.3% — build/finish quality, not any single size measurement, is the dominant Ames price driver.
- How informative is the single-feature baseline? Moderately: R^2 of 0.4589 (Dataset A) and 0.5543 (Dataset B) — in both cases, a single well-chosen feature already explains close to or over half the target variance, though far short of the multi-feature models.
- Does regularization help? Materially on Dataset B (CV RMSE improves from 36,543 to 31,615 with tuned Ridge), negligibly on Dataset A (CV RMSE unchanged at 0.7205) — directly explained by the difference in feature-space multicollinearity documented in §20 and §25.
- When do ensembles win, and what is lost? On both datasets, Random Forest or Gradient Boosting wins on raw RMSE, at the cost of coefficient-level interpretability and a materially larger train-test performance gap (§30, Table 10) — a real trade-off, not a free upgrade.
- Are errors segment-dependent? Yes on both datasets: Dataset A shows wider residual spread at high predicted prices near the census ceiling; Dataset B shows the largest residuals concentrated among a small number of very high-value sales (§30).

Underfitting, Overfitting, and Heteroscedasticity

None of the seven models underfits in an absolute sense — every model, including plain Linear Regression, comfortably beats the naive mean baseline on both datasets. Overfitting is present but bounded: Random Forest's train-test R^2 gap (0.153 on Dataset A, 0.079 on Dataset B) is the largest of any model, yet it remains the best or second-best model on held-out data, which is the expected behaviour of a well-regularised ensemble ($\text{min_samples_leaf} = 2$, 300 trees) rather than a sign the model should be rejected. Heteroscedasticity is present on both datasets — residual spread widens at higher prices — but is mild on Dataset A and more pronounced on Dataset B, tracking each dataset's target skew (§20) closely enough that the log-target ablation (§30) is a directly motivated, not speculative, response.

Multicollinearity Revisited

Dataset A's eight predictors show only moderate pairwise correlation (§20, Figure 2), and this is exactly why regularization strength barely moved the cross-validated RMSE in §25/§26 — there was little coefficient instability to fix. Dataset B's one-hot-expanded feature space, by contrast, contains genuinely correlated size-related fields (§20, Figure 9), and tuned Ridge's clear CV improvement over plain Linear Regression on that dataset is the direct, measured consequence.

32. Business Interpretation

Reading these results as a decision-support tool rather than a leaderboard changes which numbers matter most. For Dataset A's use case (a district-level valuation index), an RMSE of roughly \$50,000 ($0.5038 \times \$100,000$) from Random Forest against a \$0–500,000+ price range is a workable planning-level error band, but the identified under-prediction near the price ceiling means the model should not be trusted at face value for the highest-value districts without a manual override rule.

For Dataset B's use case (an individual-transaction appraisal), Gradient Boosting's RMSE of \$27,183 against a mean sale price of \$180,921 represents roughly a 15% typical error band — workable for a first-pass automated valuation estimate that feeds a human review step, but not precise enough, on its own, to replace an appraisal for mortgage underwriting on higher-value properties, where the residual analysis in §30 shows the largest absolute errors concentrate.

The dominant-feature findings are directly actionable: for Dataset A, a business collecting or validating home-value data should prioritise accurate location and income-proxy data over granular structural detail; for Dataset B, a business should prioritise accurate, consistently-scored OverallQual data (a subjective, appraiser-assigned rating) above any single size measurement, and should treat inconsistent quality scoring across appraisers as a material data-quality risk to the whole modelling pipeline.

Limitations

- Target censoring (Dataset A): the 1990 Census source data caps district median value near \$500,000, visibly distorting the top of the target distribution (§20, Figure 1) and degrading model accuracy exactly in that region (§30, Figure 7).
- Semantic missingness treated generically (Dataset B): fields such as PoolQC and FireplaceQu encode meaningful absence (“no pool,” “no fireplace”), but the pipeline's most-frequent imputation does not distinguish this from genuinely unknown data (§21).
- Adjusted R^2 instability on Dataset B: the classical formula breaks down when the one-hot-expanded feature count approaches the evaluation sample size, producing large negative values that should not be read as “worse than the mean” (§27).
- Elastic Net convergence warning (Dataset B): coordinate descent did not fully converge even at `max_iter = 20,000` on the wide, correlated one-hot feature space, meaning its reported coefficients may not be the exact penalised optimum (§24).
- Untuned `l1_ratio` for Elastic Net and untuned depth/estimator counts for Decision Tree, Random Forest, and Gradient Boosting: only Ridge's alpha was formally searched; the tree-based and ensemble models used fixed, reasonable-but-unoptimised settings (§25, §34).
- Random-split validity: both datasets used a random 80/20 split rather than a grouped or spatial/temporal split; Dataset A in particular has repeated nearby block groups whose geographic proximity means random splitting may understate true out-of-region generalisation error (§34).
- Small evaluation sample on Dataset B: 292 test rows and 1,168 training rows yield wider cross-validation standard deviations (§26, Table 8) than Dataset A's 4,128/16,512 split, so Dataset B's model ranking, while directionally consistent, carries more sampling uncertainty.
- No manual feature engineering: neither notebook constructs derived ratios (price per square foot, age-adjusted quality, bath-to-bed ratio) that domain literature on Ames Housing typically finds useful (§22, §34).

Future Scope

- Tune tree-depth and estimator-count hyperparameters for Decision Tree, Random Forest, and Gradient Boosting via RandomizedSearchCV, not only Ridge's alpha, to check whether the ensemble advantage widens or narrows further.
- Apply the log-target transform tested on Dataset B (§30) to the tree-based and ensemble models as well as to Ridge, and to Dataset A's target, to see whether the census-ceiling distortion in Figure 1 can be partially mitigated the same way.
- Introduce a spatial or grouped cross-validation split for Dataset A (grouping nearby block groups) and a chronological split for Dataset B (using YrSold/MoSold) to test whether performance holds under a more deployment-realistic validation protocol.
- Engineer explicit semantic-absence indicators for Dataset B's structurally-missing categorical fields (PoolQC, FireplaceQu, Alley, Fence) rather than relying on most-frequent imputation to encode them implicitly.
- Add manually engineered ratio features (price-relevant size and quality interactions) and re-run the full seven-model comparison to quantify their marginal contribution against the current, conservative feature set.
- Extend the optional XGBoost comparison referenced in the laboratory manual, alongside a computational-cost and interpretability trade-off table against Gradient Boosting's current result.
- Add explicit prediction-interval or quantile-regression output, so that deployed predictions carry an uncertainty band rather than a single point estimate, directly supporting the deployment guardrails discussed in §36.

Responsible AI

House-price models sit unusually close to questions of fairness and access, because location — the single strongest predictor on Dataset A and a strong contributor on Dataset B through Neighborhood — is itself entangled with historical patterns of segregation, redlining, and unequal public investment in many housing markets. A model that learns “location predicts price” faithfully is also, without any additional intent, learning a proxy for those historical patterns, and using it uncritically in lending, insurance, or taxation decisions risks reproducing them.

- Neither model in this study should be used as the sole basis for a credit, insurance, or tax decision about an individual or a neighbourhood; both are decision-support instruments requiring human review, consistent with §5/§23 of the source laboratory manual.
- Feature importance (Dataset B, §28) and coefficient magnitude (Dataset A, §28) describe association, not causation — a high-importance feature is not evidence that changing it would change price by the implied amount for a specific property.
- Because Latitude/Longitude (Dataset A) and Neighborhood (Dataset B) are among the strongest predictors, any operational deployment should include a fairness audit comparing prediction error and prediction bias across neighbourhoods or geographic clusters, which this academic study did not perform and explicitly recommends as a pre-deployment step.
- Both datasets are geographically and temporally bounded (California circa 1990; Ames, Iowa, sales up to 2010) and must not be assumed to generalise to other markets, price regimes, or time periods without revalidation — stated plainly rather than left implicit.

Industrial Deployment Considerations

A production deployment of either model would need to address concerns beyond what a single academic evaluation covers. The pipeline objects saved in each notebook (`house_price_pipeline.joblib`, `ames_house_price_pipeline.joblib`) already bundle preprocessing and model into one artifact accepting raw-schema input, which is a genuinely deployment-friendly design choice — it removes an entire class of train/serve preprocessing-mismatch bugs.

Concern	Current status in this study	Deployment requirement
Data drift	Not monitored; single static snapshot per dataset	Scheduled retraining and drift detection against incoming feature distributions
Prediction uncertainty	Point estimate only	Prediction intervals or quantile output before use in lending/tax contexts
Coverage checks	Not implemented	Flag predictions for properties far outside training feature ranges (extrapolation)
Fairness monitoring	Not implemented (see §35)	Neighbourhood/geography-level error-and-bias audit before go-live
Latency and scale	Not benchmarked	Random Forest / Gradient Boosting inference cost should be profiled against Ridge for high-throughput use
Versioning	<code>run_metadata.json</code> captures seed, split sizes, and selected model	Extend to full package/environment lockfile and model card

Git-hub repository Link: <https://github.com/Mohan2303-ops/House-Price-Prediction-using-Linear-Regression.git>

Conclusion

This study set out to answer how accurately, reliably, and transparently property value can be predicted from measurable attributes, using two structurally different datasets and seven model families under one shared, leakage-safe evaluation protocol. On both datasets, the answer is affirmative but bounded: every learned model comfortably outperforms a naive mean baseline (56.0% RMSE reduction on Dataset A, 69.0% on Dataset B for the best model), Linear Regression provides a stable, fully interpretable floor, and non-linear ensembles close a meaningful — though dataset-dependent — share of the remaining error, at the cost of interpretability and a larger train-test performance gap.

The extended study's specific success criteria are met: at least two structurally different datasets were used; at least five distinct model families were compared per dataset under an identical protocol (seven were actually implemented); at least one model (Ridge) was tuned via cross-validated grid search; MAE, RMSE, R^2 , and Adjusted R^2 were reported alongside residual diagnostics; train-test gaps were explicitly analysed; and a genuine ablation

— the log-target transform on Dataset B — was carried out and shown to materially improve the interpretable linear model's competitiveness against the ensembles.

The final recommendation is not simply “use the model with the lowest RMSE.” For Dataset A, Random Forest is retained as the best-performing model but the saved production artifact is a tuned Ridge pipeline, prioritising interpretability and stability given how small the ensemble's cross-validated advantage over regularized linear regression actually is once fold-to-fold variance is accounted for. For Dataset B, Gradient Boosting is the strongest raw performer, but the log-target Ridge ablation shows that most of its advantage over a well-specified linear model can be recovered without sacrificing coefficient-level transparency — the more defensible choice for a regulated valuation context, with the ensemble retained as a benchmarking companion rather than the primary decision engine.